Java1: Lesson 9 – Lab Project

1. Add **summary** at the top, **documentation** at the bottom and **comments** where necessary.
   - Recordings and interviews are an exception, but they should be included in your submission code.
2. **Cite** the source of your information.
3. Refrain from including anything **you don't fully understand**.

2

100 points

Analyze/trace the code below:

1. Add comments to Interpret the Boolean expressions in the selected if-statements (//), translating them into your own words.
2. Record your analysis by using the **debugging** mode to observe how the value of variables change and predict the outputs.

```
public class test01 {
    public static void main(String[] args) {
        TV tv1 = new TV();
        tv1.turnOn();
        tv1.setVolume(3);
        TV tv2 = new TV();
        tv2.turnOn();
        tv2.channelUp();
        tv2.channelUp();
        tv2.volumeUp();

        System.out.println("tv1's channel is " + tv1.channel + " and volume level is " + tv1.volumeLevel);
        System.out.println("tv2's channel is " + tv2.channel + " and volume level is " + tv2.volumeLevel);
    }
}
==========================================================================
public class TV {
    int channel = 1;
    int volumeLevel = 1;
    boolean on = false;

    public TV() {
    }

    public void turnOn() {on = true;}
    public void turnOff() { on = false;}
    public void setChannel(int newChannel) {
        if (on && newChannel >= 1 && newChannel <= 120) //
             channel = newChannel;}

    public void setVolume(int newVolumeLevel) {
        if (on && newVolumeLevel >= 1 && newVolumeLevel <= 7) //
             volumeLevel = newVolumeLevel;}

    public void channelUp() {
        if (on && channel < 120) //
            channel++;}

    public void channelDown() {
        if (on && channel > 1)
            channel--;}

    public void volumeUp() {
        if (on && volumeLevel < 7)
            volumeLevel++;}
```

```
    public void volumeDown() {
        if (on && volumeLevel > 1) //
            volumeLevel--;}
}
```

Sample output	
File(s) to submit	J109_2.java YouTube link

3

100 points

Write a test code to calculate the areas of circles with radii of = 1, 10, 100:

1. Utilize the `CircleWithPrivateDataFields` class: https://liveexample.pearsoncmg.com/html/CircleWithPrivateDataFields.html
2. Submit only one .java file that includes the class.
3. Record how you analyze your code using debugging mode.

**Make sure that your implementation is different from the one provided.*

Sample output	
File(s) to submit	J109_3.java YouTube link